Element G

Construction Of A Testable Prototype

The team made a 3D model with exact measurements for the hydroponics unit, and it featured many desired functions that were given by the team and our stakeholders. Additionally it featured things the team had thought of when researching for this project. Our prototype will be a hydroponics unit that fits into the measurements 30” by 60” that features waterproof materials, multiple sections for plants, multiple trellises, a working pump and light controlled by a single panel, and is easy to move around by a single person. Thus our design included those aspects while adding in pH testers and added storage to make nutrient solutions easily accessible to our stakeholders. The method that the team will use to ensure that the hydroponics will meet all the requirements is by collaborating with each other and pitching different ideas to experts to get their opinions. With that the team will reiterate and repeat the process over again until our team can get a solution that satisfies all requirements. Our team will test viability by presenting it to experts and our stakeholders to see if there are any alterations needed and any information that is missing. One thing from our design that was missing was a representation of our chosen pump which made it confusing for our audience to understand the design. Our team can use the review form Mr. Roberson to make alterations which justifies our decisions to change aspects that relate to the water basin and our trellis. Our design solution makes use of technology and research to accomplish a prototype on a sketching app that displays the plan to build a future hydroponics design. With this design the team is able to continue on a future project to provide a workable prototype for our stakeholder. To make our prototype waterproof in our design the team has chosen weather shield pine lumber to maintain our aesthetically pleasing design while it staying functional. To help us space our plants apart our team has chosen PVC pipes as our secondary structure that holds the actual plants that would grow in the 3D Model. The team will test the viability of our design with regard to comparison by measuring the designated area with the goal of our 3D designs measurements matching as a minimum for success. Our team has multiple measurements from the sketching software that allow us to properly construct this project in a future task. Despite not being able to provide more traditional data points, we believe that material use is a crucial factor to success of the solution because even though we are not building. Our team knowing the materials open to us financially will allow us to go through the project smoothly despite the fact no building will be done. Our design was reviewed by James Roberson who stated our design needed some alterations for it to be successful. Our team has shown him with the hope of him stating any falters and providing us suggestions on how we as a team could improve our prototype so that it is ready to present to our stakeholders. In order to complete our 3D model prototype we will need to look back at our designs measurements, and features, which will address any concerns that were brought up by Mr. Roberson during our meeting. This will also allow us to give our stakeholders a final design that addresses all their consumer needs.